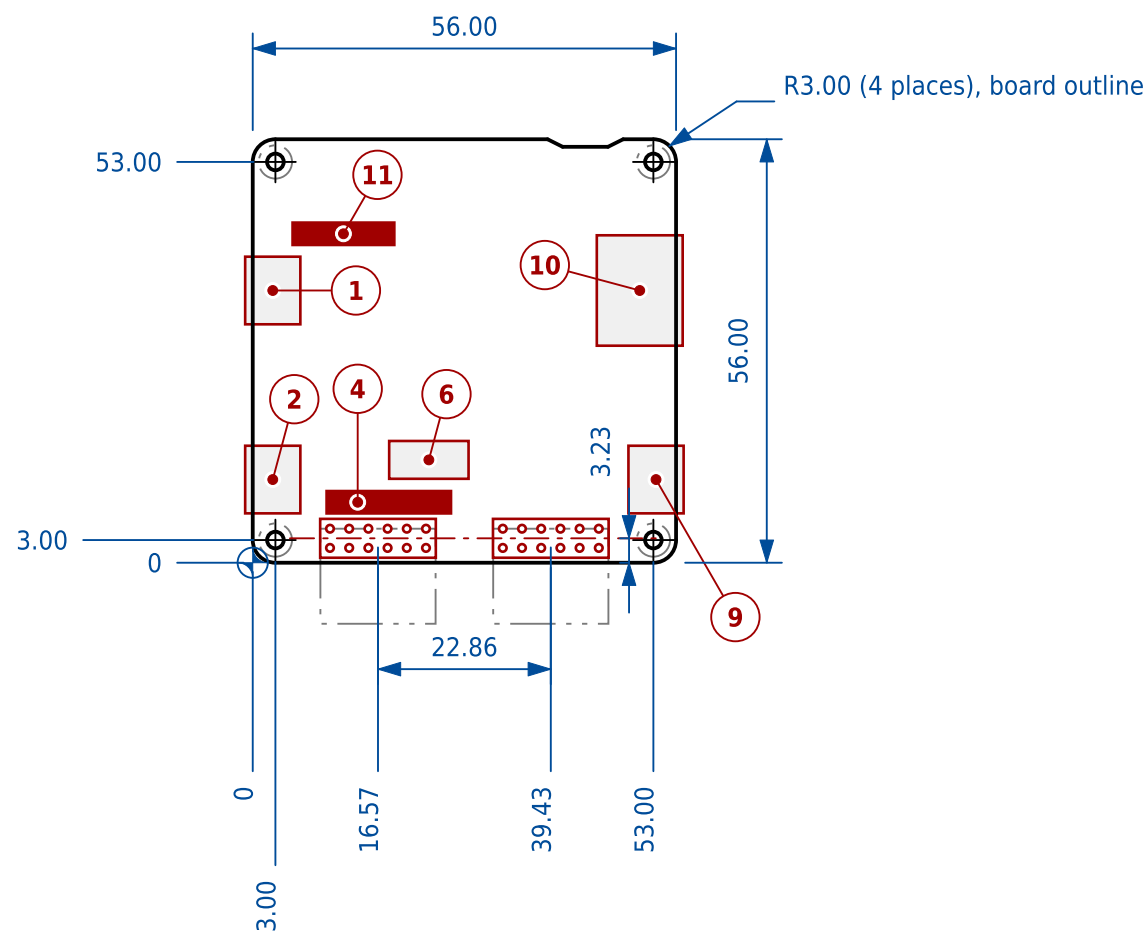




NOTES

1. Viewed from the component side.
2. Every dimension is read from the board file; Great Scott Gadgets publish no mechanical drawing. Hole IDs are the KiCad reference designators.
3. Schedule numbers 9, 10 and 11 were added to the family for this board. 1 to 8 mean what they mean on the other FPGA sheets, so the four USB ports read 1, 2, 9, 10.
4. No Ethernet jack.
5. Feature 6, J5, is a surface-mount mezzanine receptacle on the component side: a board-to-board expansion socket well inside the outline, not an edge port like the expansion connectors on the other sheets.
6. Pmod A and B are right-angle sockets: the pin field is on the board and the housing hangs off the front edge, so a peripheral plugs in level with the board rather than standing up from it. The housings reach 8.07 mm past the edge.
7. The board ships inside an enclosure. The outline drawn is the bare PCB's; the repository publishes no case dimensions.
8. Buttons PROGRAM and USER on the left edge and RESET on the right are not drawn. They are side-actuated tactile switches whose bodies reach 2.50 mm past the left edge and 2.50 mm past the right.
9. Assembled envelope, connector overhang and the Pmod housings included: 58.00 x 64.07 mm. With the three side buttons as well it is 61.00 x 64.07 mm, which is what a case has to clear.

NOTES (continued)

10. The top edge carries a recess 1.00 mm deep, flat from $x = 41.00$ to 47.00 , each end ramping 2.00 back to the edge at $x = 39.00$ and 49.00 . The board file gives no purpose for it.

SOURCES

1. KiCad board file:
<https://github.com/greatscottgadgets/cynthion-hardware>
cynthion.kicad_pcb @ 13aa71c2 (tag r1.4.0) - title block:
Cynthion rev 1.4.0, Copyright 2019-2024 Great Scott
Gadgets, resolved from the project file's text variables at the
same commit; Great Scott Gadgets.
2. Revision sold:
[https://github.com/greatscottgadgets/cynthion-hardware/
releases/tag/r1.4.0](https://github.com/greatscottgadgets/cynthion-hardware/releases/tag/r1.4.0) - the r1.4.0 release notes read "Initial
production release."; it is the newest release and the tip of
the repository, and no commit since has touched the board
file.
3. Licence:
[https://github.com/greatscottgadgets/cynthion-hardware/
blob/r1.4.0/LICENSE](https://github.com/greatscottgadgets/cynthion-hardware/blob/r1.4.0/LICENSE) - CERN Open Hardware Licence Version
2 - Permissive, Copyright (c) 2019-2024 Great Scott Gadgets.
4. Ports and indicators:
[https://cynthion.readthedocs.io/en/latest/hardware/
device_overview.html](https://cynthion.readthedocs.io/en/latest/hardware/device_overview.html) - names the four USB ports CONTROL,
AUX, TARGET C and TARGET A, in that order; six user LEDs
driven by the FPGA and five status LEDs driven by the debug
microcontroller.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	3.00	3.00	2.20	4.40
H2	3.00	53.00	2.20	4.40
H3	53.00	53.00	2.20	4.40
H4	53.00	3.00	2.20	4.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
Pmod A	J7	bottom	16.570	3.230	22.920	4.500
Pmod B	J8	bottom	39.430	3.230	45.780	4.500

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J2, CONTROL port	-1.00 to 6.30	31.53 to 40.47
2	USB-C J1, AUX port	-1.00 to 6.30	6.53 to 15.47
3	Ethernet RJ45 - not on this board	-	-
4	User LEDs D2-D7, 6 on 3.00 mm pitch	9.77 to 26.23	6.52 to 9.48
5	User LEDs, tri-colour - not on this board	-	-
6	Mezzanine receptacle J5, 30-way	18.05 to 28.55	11.10 to 16.10
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-
9	USB-C J4, TARGET C port	49.70 to 57.00	6.53 to 15.47
10	USB-A J3, TARGET A port	45.52 to 56.88	28.70 to 43.30
11	Status LEDs D10-D14, 5 on 3.00 mm pitch	5.27 to 18.73	42.02 to 44.98

LEGEND

- | | |
|---|------------------------------------|
|  | Board outline |
|  | Component body, scheduled |
|  | Pmod connector body, hole keep-out |
|  | Dimension, extension and leader |

CYNTHION

r1.4.0

DRAWN generated		VERSION v0.0-283-g0a10fb4	UNITS mm	SCALE 1:1
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-CYNTHION		REV A
MATERIAL PCB, 1.6 nominal		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08		